DETECTING NONZERO COHOMOLOGY OF A PSEUDO-COHERENT COMPLEX BY A MAP TO A FINITELY GENERATED ANNIHILATED MODULE

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Allowed standard prerequisites (editorial).

- Equivalent formulation of pseudo-coherence as finite cohomology in bounded degrees. Lemma \ref{lemma-Noetherian-pseudo-coherent}, cited at proof line 16835. Reconciles the formulation used in the statement.
- Existence of a detecting map from a pseudo-coherent complex to a shift. Lemma \ref{lemma-detect-cohomology-pseudo-coherent}, cited at proof lines 16836-16837. Provides the map that the Artin-Rees step refines.
- Artin-Rees: the intersection of a submodule with a large power of the ideal lies in the product. Algebra, Lemma \ref{algebra-lemma-Artin-Rees}, cited at proof lines 16838-16840. Chooses the power used to define the detecting module.

Source reference. Original lemma 0A7D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12876–12876. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 066E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7933–7933. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0G8Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20283–20283. Navigation pointer to a statement the

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Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Noetherian ring. Let $K \in D(A)$ be pseudo-coherent, i.e., $K \in D^-(A)$ with finite cohomology modules. Let $\mathfrak{m}$ be a maximal ideal of A. If $H^i(K)/\mathfrak{m}H^i(K) \neq 0$, then there exists a finite A-module E annihilated by a power of $\mathfrak{m}$ and a map $K \to E[-i]$ which is nonzero on $H^i(K)$.*

Proof. (The equivalent formulation of pseudo-coherence in the statement of the lemma is Lemma 066E.) Choose $K \to M[-i]$ as in Lemma 0G8Y. By Artin-Rees (Algebra, Lemma 00IN) we can find an n such that $H^i(K) \cap \mathfrak{m}^n M \subset \mathfrak{m}H^i(K)$. Take $E = M/\mathfrak{m}^n M$. □